R_MU_Apt_HR_pre1945

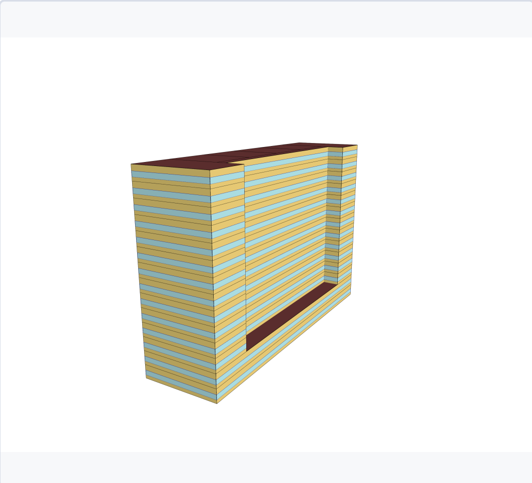
VERSION | DATE
2026-06-15

SECTION 1 — MODEL DESCRIPTION

BUILDING DESCRIPTION

Sector	Residential
Building type	Multi-Unit
Building subtype	Apartment - HR
Vintage	A_Pre1945

3D MODEL



Three-dimensional geometric representation of the building model used for energy simulation and performance analysis.

BUILDING CHARACTERISTICS

Floor Area [m²]	13792
Climate zone	6A

HVAC SYSTEM

Cooling system	Central Electric Heat Pump System
Heating system	Central Heat Pump / Boiler
Ventilation system	Corridor Exhaust + Fresh Air Make-Up
Typical COP cooling	3.50 – 5.00

INSULATION & AIR TIGHTNESS

Exterior walls [W/m²/K]	1.5
Interior walls [W/m²/K]	2.51
Roof [W/m²/K]	2.49
Slabs [W/m²/K]	1.13
Infiltration [m³/h/m²]	18.0

GLAZING

Glazing [W/m²/K]	3.32
SHGC	0.66

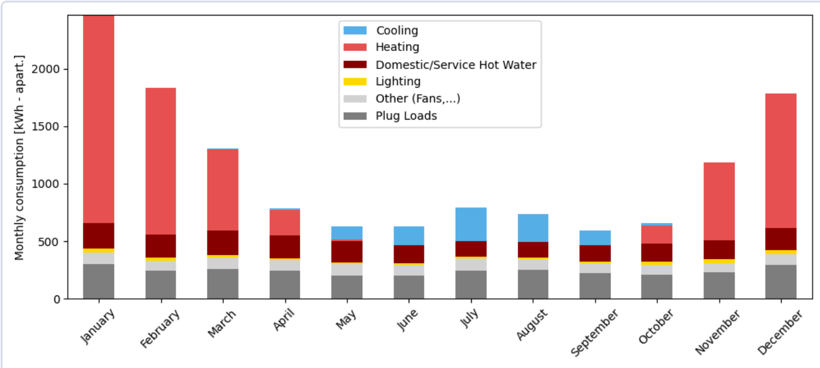
WINDOW-TO-WALL RATIO

Average [%]	35.0

ANNUAL END USES — ELECTRICITY [kWh]

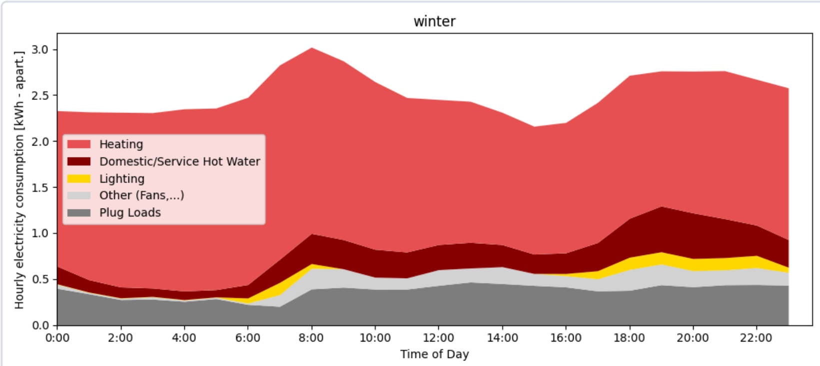
END USE	[kWh/year per apart.]
Heating	6054
Cooling	977
Interior Lighting	252
Exterior Lighting	17
Interior Equipment	4993
Fans	92
Pumps	1
Water Systems	1023
Total End Uses	13410

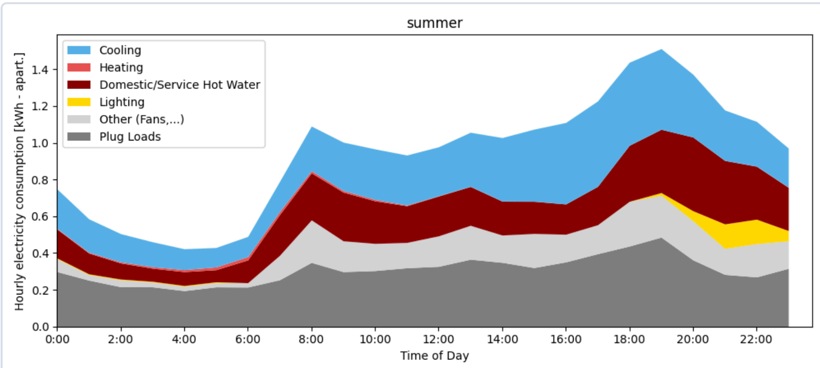
MONTHLY CONSUMPTION BY END USE



Monthly energy consumption breakdown by end use, showing the contribution of heating, cooling, lighting, domestic hot water, plug loads, and auxiliary systems throughout the year.

AVERAGE DAILY PROFILE BY USAGE





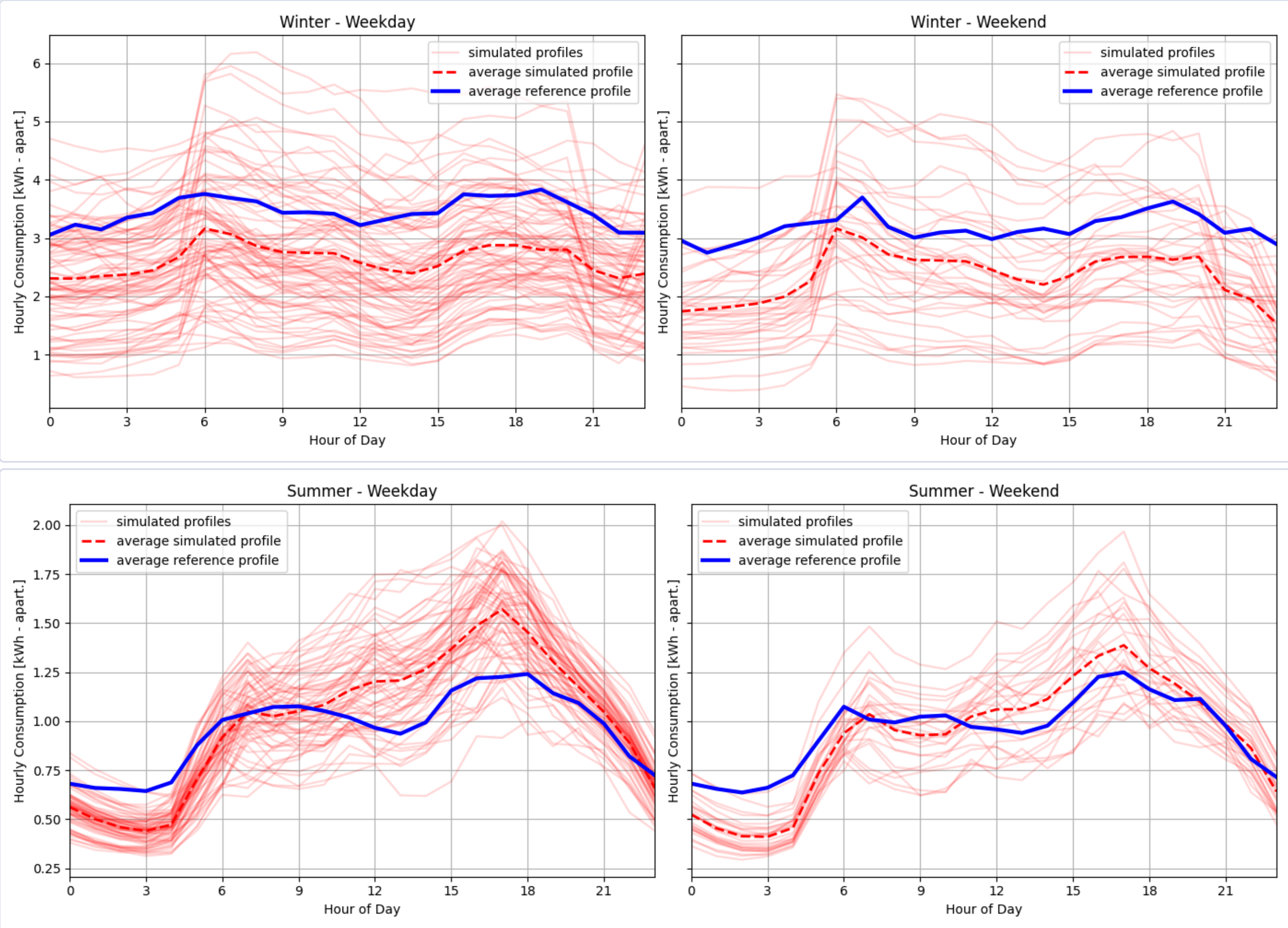
Average daily consumption profile for winter and summer, showing the contribution of cooling, lighting, domestic hot water, plug loads, and auxiliary systems throughout an average winter and summer day.

ENERGY USE INTENSITY	TOTAL CONSUMPTION	ELECTRICITY	NATURAL GAS
1	13410	13410	0
kWh/m²/yr	kWh/yr	kWh/yr	kWh/yr

DONNÉES DE RÉFÉRENCE

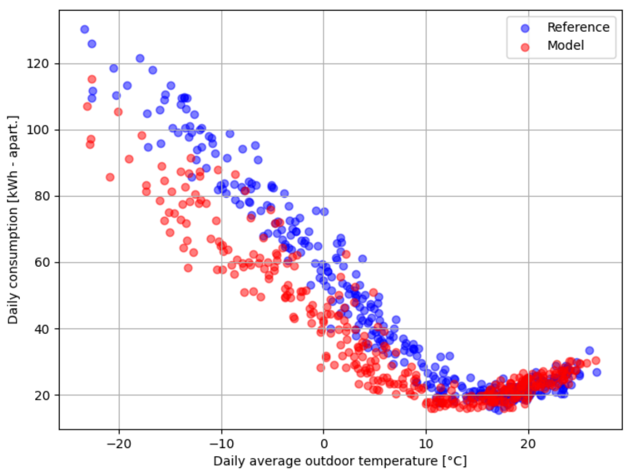
The reference data used for the comparison come from a dataset of more than 60,000 residential electricity meters, combined with metadata. Filters were applied to ensure consistency with the building type and subtype of this card, as well as to exclude end uses such as pool, spa, and electric vehicle charging.

DAILY PROFILE



Daily consumption profiles for summer and winter comparing simulated hourly energy use with the reference data. Red lines represent individual days from the model for the given period, the red dashed line indicates the average simulated profile, and the blue line shows the average profile of the reference data. The figures on the left represent weekdays, while the figures on the right represents weekends.

DAILY CONSUMPTION VS. DAILY AVERAGE OUTDOOR TEMPERATURE



Comparison between modeled daily consumption and reference consumption as a function of outdoor temperature, for all days of the reference year.

COMPARISON TABLE — KEY INDICATORS

INDICATOR	MODEL	REFERENCE	DELTA
Yearly electricity consumption [kWh]	13410	13410	13410
Baseload consumption [kWh/day]	18.1	20.4	-2.3
Electric heating slope [W/K]	-104.7	-127.1	22.4
Cooling slope [W/K]	43.8	43.2	0.6
Winter peak demand - AM [kW]	6.2	6.8	-0.6
Winter time of peak - AM	8.0	6.0	2.0
Winter peak demand - PM [kW]	5.6	6.5	-1.0
Winter time of peak - PM	17.0	20.0	-3.0